

Multiplicative order

 12 languages 

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In [number theory](#), given a positive integer n and an [integer](#) a [coprime](#) to n , the **multiplicative order** of a modulo n is the smallest positive integer k such that $a^k \equiv 1 \pmod{n}$.^[1]

In other words, the multiplicative order of a modulo n is the [order](#) of a in the [multiplicative group](#) of the [units](#) in the [ring](#) of the integers [modulo](#) n .

The order of a modulo n is sometimes written as $\operatorname{ord}_n(a)$.^[2]